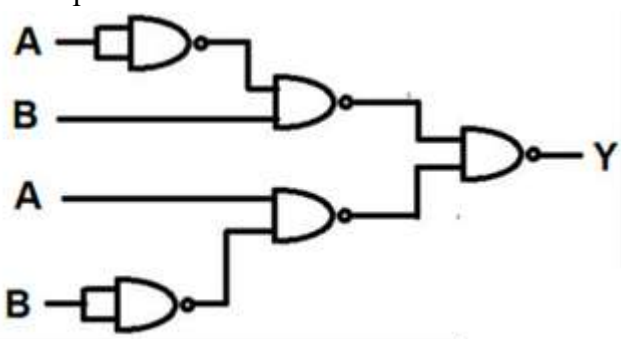


Qn No.	Tutorial Questions-Combinational Circuits	Marks	CO
1	Do the conversion to the following representation: a) $(3A9E.B)_{16}$ to binary and decimal b) $(110101101)_2$ to corresponding Gray code. c) $(-107)_{10}$ into signed 12-bit number in the i) Sign Magnitude ii) 1's complement iii) 2's complement form	10	CO1
2	Using a 4-variable Karnaugh Map (K-Map), simplify the below function and realize using NAND gates only. $f(A, B, C, D) = \sum (1, 4, 9, 10, 11, 12, 14) + d(0, 8, 13)$	10	CO2
3	Analyse the logic circuit diagram shown below, a) Derive the simplified Boolean expression at the output Y and support your answer with appropriate Boolean reduction steps. b) If all NAND gates in the given circuit are replaced with NOR gates, derive the Boolean expression at the output Y. 	10	CO1
4	Using a standard 4-bit Adder/Subtractor unit, design a circuit that performs an increment-by-one and decrement-by-one operation on a 4-bit binary input.	10	CO3
5	Implement the minimal cost POS expression using K-Map for the function given below using NOR gates only. a) $f(A, B, C, D, E) = \prod (0, 1, 2, 4, 7, 8, 12, 14, 15, 16, 17, 18, 20, 24, 28, 30, 31)$ b) $f(A, B, C, D) = \prod (0, 2, 4, 5, 6, 7, 8, 10, 12, 13, 14, 15)$	10	CO2
6	Find the minimal cost SOP expression for the function given below and implement logic circuit using NAND gates only. $f(A, B, C, D, E) = \sum (0, 2, 4, 7, 8, 10, 12, 16, 18, 20, 23, 24, 25, 26, 27, 28)$	10	CO2
7	A new Binary Coded Senary (BCS) number system is proposed, in which each digit of a base-6 number is represented by its corresponding 3-bit binary code. For instance, the base-6 number $(24)_6$ is represented in BCS as 010100, where each digit is encoded using 3-bit binary. Given the BCS code 100010011001, determine the corresponding number in the base-6 number system.	2	CO3
	Convert the decimal number $(501)_{10}$ into its equivalent representation in the base-6 number system.	3	

	Convert the decimal number (501) ₁₀ into its equivalent representation in the binary code and gray code.	5	
8	a) Design a combinational logic circuit which do the bit-wise addition in binary arithmetic, accounting for both the current bits and any carry from prior computations. Implement the given circuit using 2-to-4-line decoders and any necessary external logic gates.	5	CO3
	b) For the function $f(A,B,C) = \sum m(1,4,5,7)$, use Shannon's expansion theorem to derive an implementation using a 2-to-1 multiplexer and any other necessary gates.	5	
9	A chemical storage tank is monitored using four sensors that measure temperature (T), pressure (P), humidity (H), and pH level (X) in this order. Each sensor's interface circuit outputs a logic HIGH when its respective parameter exceeds a critical threshold. For safety compliance, an alarm system that is triggered by a logic LOW input must activate whenever either temperature and pressure become critical, or when both humidity and pH simultaneously exceed their limits. Design a circuit for this application using basic gates.	10	CO1
10	a) Simplify the Boolean expression $Y = AB + A(B+C) + B(B+C)$ and draw a circuit with basic gates. i. Implement the reduced function with NAND gates only. ii. Implement the reduced function with NOR gates only.	1+2+2	CO1
	b) A, B, and C are the decimal integers corresponding to the 4-bit binary number 1010 considered in signed magnitude, 1's complement, and 2's complement representations, respectively. Find out the 6-bit 2's complement representation of (A+B+C).	5	CO1
11	a) Design a Binary to Gray code converter and Gray to Binary Converter b) Design a binary to BCD converter	10	CO3
12	a) Design a 4-to-2 priority encoder. With the help of proper table, derive the output expressions through K-map simplification and implement the final circuit using only with 2-input basic logic gates. Additionally, find out a real-time application where such a priority encoder would be effectively used.	5	CO3
	b) Design the given Boolean function using a 3×8 decoder whose outputs are active LOW and 2 input logic gates. $f = \sum m(0,2,4,5,7)$	5	
13	Design 4-bit binary to gray code converter. Show the truth table and implement the circuit using AOI circuit.	10	CO3
	Synthesize the below function, using $f(A,B,C,D) = \sum (1,2,4,7,9,10,11,12,14)$ c) 8x1 MUX d) 4x1 MUX 3x8 decoder	10	CO3
14	Find the cost and minimal SOP function. Realize using NAND alone circuit. a) $F(A,B,C,D) = \sum m(0,1,2,5,7,8,9,10,13,15)$	40	CO2

	b) $F(A, B, C, D) = \Sigma m(0, 1, 3, 5, 7, 8, 9, 11, 13, 15)$ c) $F(A, B, C, D) = \Sigma m(1, 3, 4, 6, 8, 9, 11, 13, 15) + \Sigma d(0, 2, 14)$ d) $F(A, B, C) = \Sigma m(0, 1, 6, 7) + \Sigma d(3, 5)$ e) $F(A, B, C) = \Sigma m(1, 2, 5, 7) + \Sigma d(0, 4, 6)$ f) $F(A, B, C, D) = \Sigma m(0, 2, 8, 10, 14) + \Sigma d(5, 15)$ g) $F(A, B, C, D) = \Sigma m(3, 4, 5, 7, 9, 13, 14, 15)$ h) $F(W, X, Y, Z) = \Sigma m(1, 3, 4, 6, 9, 11, 12, 14)$		
--	---	--	--

